

Interpretable Semantic Medical Image Segmentation With Style and Confidence

Wei Dai , Siyu Liu , Jurgen Fripp , Craig Engstrom , and Shekhar S. Chandra , *Member, IEEE*

Abstract—The scarcity of semantically labelled data presents major challenges for medical image segmentation using deep learning models, and the “black-box” nature of these models inherently limits their interpretability during clinical deployment. To address these issues, we propose Generative Adaptable Segmentation Evolution (GASE), an end-to-end, style-based generative adversarial framework for robust and interpretable medical image segmentation under extreme data scarcity. Trained on limited single-sequence Magnetic Resonance (MR) data, GASE self-adapts to unseen acquisition-level image variations—arising from differences in scanning protocols, imaging sequences, and patient demographics—while simultaneously improving interpretability through automatic input validity assessment and output reliability estimation. During adversarial training, the style-learning generator captures a manifold-like representation of input style features for valid input interpretation and diversifies labelled training pairs through style interpolation to simulate acquisition-level image variations. The segmentation-based discriminator, trained on these diversified samples, further improves adaptation to unseen styles and automatically estimates the reliability of predicted masks via confidence learning. Extensive experiments on knee and pelvis MR image datasets demonstrate that GASE achieves high segmentation accuracy for assessments of multiple tissues under examination, strong adaptability to unseen acquisition-level image variations, and intrinsic interpretability, underscoring its potential for reliable clinical deployment.

Index Terms—Automated medical image analysis, semantic segmentation, GAN, model interpretability, confidence learning, MR, style-based learning.

I. INTRODUCTION

RECENTLY, Deep Learning (DL) models, such as Convolutional Neural Networks (CNN) and transformers, have become major research topics for automatic image segmentation. Transformers, which focus on long-term dependencies using self-attention mechanisms [1], have shown promising results on various natural image segmentation problems [2], [3], especially when pre-trained on extremely large image datasets.

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In comparison, CNN [4], [5], [6] generally perform better on limited datasets due to their hierarchical structures for feature extraction and inductive bias for local connectivity, and have been widely proposed for medical image segmentation tasks [7], [8], [9], [10]. The efficient computation of CNN also makes them well-suited for clinical imaging environments with limited computing resources.

A common challenge in medical image segmentation using DL models, especially for MR images, is the scarcity of labelled training samples, largely due to the labour-intensive nature of manual annotation and the restricted data access imposed by privacy regulations. This limitation often leads to model overfitting, where models perform well in controlled testing scenarios but fail to generalise in real-world clinical applications. A key contributor to this issue is the acquisition-level variability in MR images, arising from differences in scanner vendors, magnetic field strengths, patient demographics, and imaging sequences. Unlike contrast-level variations between T1- and T2-weighted MR images or modality-level differences between MR and CT—which introduce substantial changes in tissue contrast—acquisition-level image variations are more subtle yet frequently encountered, as models are typically tested on data from the same contrast used during training. These variations occur within a consistent physical contrast mechanism and primarily affect factors such as signal intensity and signal-to-noise ratios. Therefore, there is a pressing need for an automated segmentation framework that achieves state-of-the-art performance while remaining robust to acquisition-level image variability under limited training data conditions, thereby enabling reliable clinical deployment.

To address variability under data scarcity, semi-supervised and unsupervised methods [11], [12], [13] have been proposed to leverage unlabelled data or multi-sequence MR datasets for enhanced segmentation performance. Multi-modal generative approaches [14], [15], [16] have also been explored to augment labelled medical images using standalone augmentation models. However, these methods are often constrained by the availability of additional imaging datasets or rely on artificially generated label maps during mask augmentation, which may compromise their output reliability in clinical settings. Moreover, their multi-model architectures and multi-stage pipelines introduce considerable deployment complexity, posing challenges for non-AI-expert clinical users. Therefore, an automated single-stage, user-friendly segmentation framework that robustly handles both data scarcity and acquisition-level image variability is highly desirable for practical clinical deployment.

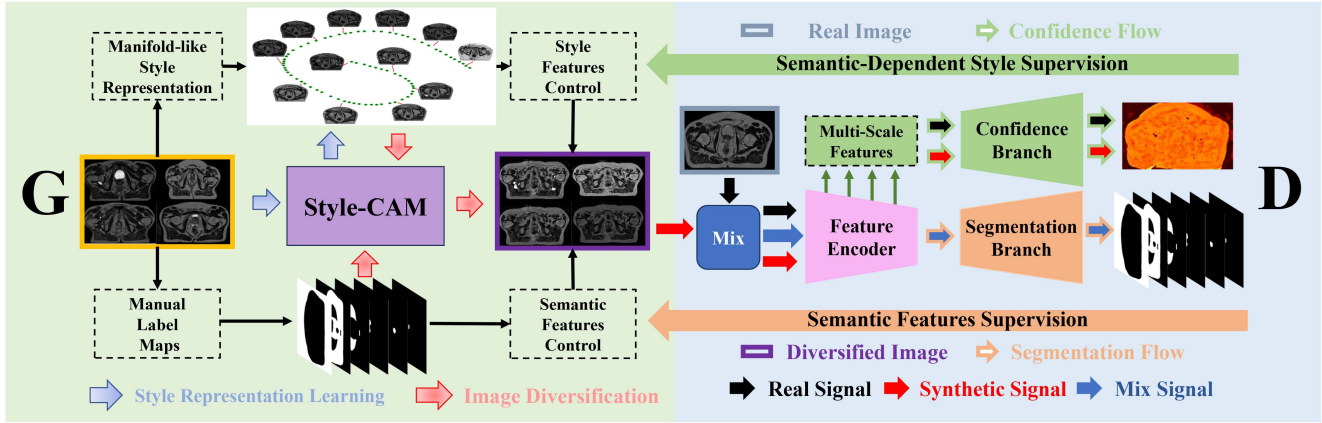


Fig. 1. Overview of GASE's single-stage end-to-end pipeline with pelvic axial MR image examples. The generator (G) performs both image diversification and input interpretation via manifold-like representation learning of structure-dependent style features derived from training samples. The discriminator (D) is responsible for image segmentation and is trained on the mixed samples for better adaptation to unseen variations and simultaneously assesses output reliability through confidence learning based on multi-scaled features extracted from segmentation branch.

To adapt DL-based segmentation models for acquisition-level image variability while maintaining an user friendly pipeline, NNUNet [9] was proposed to automatically configure a UNet-based [7] segmentation pipeline for optimal performance on a given image dataset. As a self-adapting framework, NNUNet employs an extensive pre-processing and augmentation pipeline to improve robustness under data scarcity and variability without the need for task-specific tuning. However, its self-configured pre-processing pipeline is not fully data-driven, relying instead on grid-search and pre-defined heuristic rules. Moreover, the resulting models often have a large memory footprint and slow inference speed due to the pipeline's complexity, making NNUNet less suitable for clinical deployment where hardware resources are frequently constrained. Furthermore, NNUNet provides no mechanism for interpreting its learned input or output spaces after training. As DL models are prone to hallucination and can behave unpredictably when encountering inputs outside the training distribution, the lack of interpretability limits NNUNet's reliability in safety-critical medical settings.

In this study, we introduce Generative Adaptable Segmentation Evolution (GASE), an automated semantic segmentation framework designed to simultaneously address data scarcity, acquisition-level image variability, and model interpretability, while ensuring user-friendly clinical deployment. An overview of the proposed GASE framework is illustrated in Fig. 1, and the main contributions are summarized as follows:

- 1) To enable an end-to-end pipeline that is accessible to end-users, GASE integrates both image synthesis and segmentation networks within a single-stage feed-forward Generative Adversarial Network (GAN). The framework is fully supervised with a limited single-sequence MR dataset, thereby eliminating the need for additional image datasets or artificially generated masks required by other semi- or unsupervised approaches.
- 2) To enhance the robustness of the discriminator (D), which performs the segmentation task, against acquisition-level image variations, the generator (G) unifies conventional preprocessing and augmentation stages into a data-driven

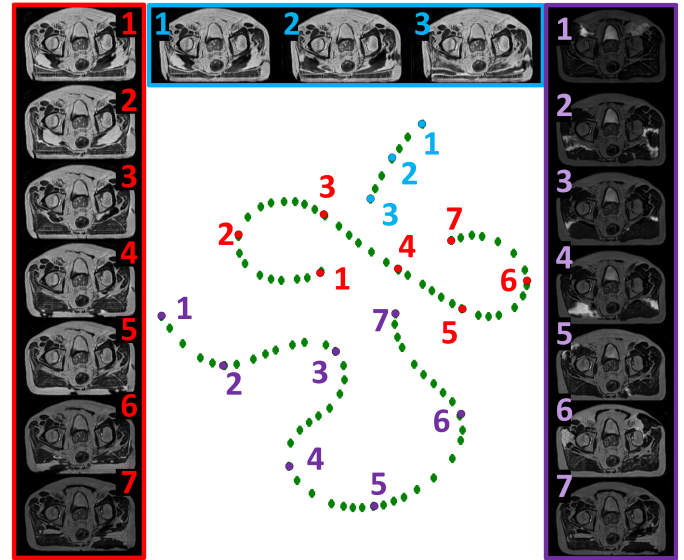


Fig. 2. Learned manifold-like representation of semantic-dependent style features from the pelvis MR dataset. Each green dot represents a projected style vector from the representation, and combining these vectors with a fixed label map produces the diversified images shown on the side. The manifold-like representation reveals consistent semantic structure with varying style attributes such as local intensity and global contrast.

image synthesis module. This G automatically diversifies the appearance (style) of the labelled image pairs while preserving their structural integrity as ground-truth annotations during training.

- 3) Regarding model interpretability, the G learns a manifold-like representation of input style features (shown in Fig. 2) through style-based learning, which allows identification of valid segmentation inputs for the trained D . Meanwhile, the D incorporates confidence learning to automatically assess the reliability of its predicted outputs without human intervention.
- 4) Unlike conventional feature diversification strategies, in which the generative process is independent of the

segmentation task during training, our GASE framework integrates both tasks within a unified architecture. This integration enables the segmentation module to actively provide confidence feedback to the generative network through confidence learning, thereby guiding it to produce anatomically consistent and task-relevant features rather than relying solely on input samples.

- 5) Evaluated on three MR datasets of the pelvis and knee with limited training samples, our GASE achieves state-of-the-art image segmentation performance under consistent acquisition conditions and demonstrates superior generalisation when confronted with unseen acquisition-level image variations during inference. Its end-to-end, feed-forward pipeline—featuring uniquely interpretable inputs and outputs—distinguishes it from existing generative models, highlighting its potential as a reliable tool for medical image segmentation, particularly in clinical settings where understanding the model’s decision-making process is essential.

II. RELATED WORKS

Direct Segmentation: Methods with encoder-decoder architectures have been widely used for automated medical image segmentation [8], [17], [18], [19], [20], reconstructing label maps based on abstract features extracted by encoders. To avoid information loss during the encoding process while preserving sufficient receptive field, dilated convolution [21] has also been applied, replacing down-sampling operations for fast and efficient medical image segmentation [22], [23], [24]. GAN-based methods [25], [26], [27], [28], [29], known for high-quality cross-domain image translation, have also been proposed to translate medical images into label domain using a G with a D supervising the quality of predicted labels. Semi- or weakly supervised methods have also been proposed to leverage multi-modal or unlabelled data for direct medical image segmentation [30], [31], [32]. Unlike most existing approaches, our GASE simultaneously performs mask-to-image and image-to-mask translations using G and D , respectively, trained in an adversarial manner.

Augmentation-based Segmentation: While direct segmentation approaches can achieve high accuracy, they are often vulnerable to overfitting due to data scarcity and acquisition-level image variability. A common strategy to mitigate this issue is data-driven augmentation [33], [34], [35], which expands the labelled dataset through synthetic image generation. Some methods further rely on artificial label maps that are translated into target image domains to create additional training pairs [36], [37], [38], [39]. However, these approaches typically require multi-stage pipelines involving separate augmentation and segmentation models, making deployment cumbersome—particularly for non-AI-expert users. Single-model frameworks that integrate augmentation and segmentation end-to-end [26], [40] have also been explored, but they often require multiple pairs of GANs, substantially increasing architectural complexity and computational cost. In contrast, our GASE framework provides an automated segmentation solution with an online

diversification mechanism that enables self-adaptation to unseen image variations. By jointly performing image diversification and segmentation within a single feed-forward adversarial pipeline, GASE eliminates multi-model or multi-stage training, thereby offering a more streamlined and deployment-friendly solution for clinical use.

Confidence-based Segmentation: In conventional GANs, the D performs binary classification and acts as a high-order spatial consistency loss for improving the G . However, this provides only a global confidence score, neglecting local reliability and often yielding suboptimal supervision when training G for image segmentation task. To address this limitation, confidence learning [41], [42] introduces 2D probability maps that reflect the similarity between G ’s predictions and ground-truth masks on pixel-wise, and has been adopted within D for semi-supervised medical image segmentation [43], [44], [45]. Yet directly assessing the “realness” of predicted masks can be inefficient due to the highly abstract nature of semantic labels. Instead, our D performs confidence learning by computing semantic-dependent, pixel-wise confidence map based on multi-scale features extracted directly from the input image during segmentation. This measures how closely an input aligns with the training distribution in both style and semantic characteristics, enabling reliable detection of uncertain or erroneous outputs when inputs fall outside the learned domain.

Style-based Generative Methods, originally developed for style transfer, have recently been adopted for augmentation-driven image segmentation [46], [47], [48], [49], [50], [51]. While effective at capturing style features for augmentation, these approaches generally lack the ability to store or interpret the underlying style feature space—an essential requirement for trust and reliability in medical imaging applications. Style-GANs [52], [53] address this by learning style representations through a mapping network that embeds input styles into an intermediate latent space. Several StyleGAN-based methods [54], [55], [56], [57], [58], [59] have been explored for medical image segmentation and classification, but none fully exploit the learned latent space for input interpretation or provide confidence estimation for their outputs. In contrast, our framework integrates the mapping network within G with confidence learning from D in an adversarial manner, enabling the latent space to encode semantic-dependent style features derived from the training data. This latent space not only supports interpretable input validation but also guides output reliability assessment through confidence learning, providing a unified, style-aware mechanism for robust and trustworthy segmentation.

III. METHOD

A. Preliminaries

Generative Adversarial Network (GAN) [60] consists of two sub-networks: a G and a D , which engage in a min-max game during training. The G attempts to deceive the D by approximating a distribution, p_g , that closely mirrors the real data distribution, p_r . Meanwhile, the D is trained to distinguish between p_r and p_g . Since the D acts as a binary classifier and provides a “variational” loss for the G , a min-max loss based

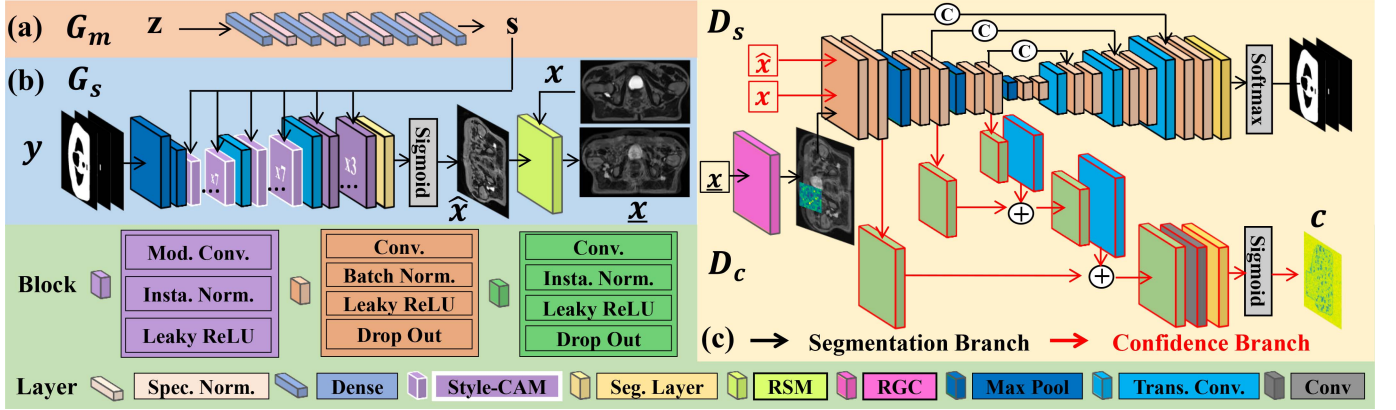


Fig. 3. Detailed architecture of GASE illustrated with pelvic axial MR images. (a) The mapping network G_m maps noise z to a style vector s . (b) The generator G_s combines label map y with s to produce diversified sample $\hat{x}$; the mixed signal $\underline{x}$ is formed by blending $\hat{x}$ with real image x via Real Synthetic Mixup (RSM). (c) The discriminator's segmentation branch D_s (black) processes $\underline{x}$ using Random Gaussian Cutout (RGC) and outputs the predicted mask $\hat{y}$, while the confidence branch D_c (red) performs confidence learning and produces the confidence map c .

on cross-entropy is typically used to optimise both models, given by:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_r(x)} [\log(D(x))] + \mathbb{E}_{x \sim p_g(x)} [\log(1 - D(x))], \quad (1)$$

where $\mathbb{E}$ denotes the expected value over the distributions, and $D(\cdot) \in [0, 1]$ represents the likelihood that the input is real. The $\min_G \max_D$ formulation indicates that the loss is minimised for the G and maximised for the D . For the optimal D solution $D^*(x) = \frac{p_r(x)}{p_r(x) + p_g(x)}$, which maximises (1), the G 's objective becomes:

$$\min_G V(D^*, G) = 2D_{JS}(p_r \| p_g) - 2 \log 2, \quad (2)$$

showing that minimising the G 's loss is equivalent to minimising the Jensen-Shannon divergence between p_r and p_g , guiding the G to produce samples that resemble real data.

Manifold Mapping: For a high-dimensional dataset $X = \{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^D$, each data point x_i is hypothesised to lie on a smooth manifold M , embedded within $\mathbb{R}^D$, but with a lower intrinsic dimension d (where $d \ll D$). This relationship can be expressed as: $f: M \subset \mathbb{R}^d \rightarrow \mathbb{R}^D$. By learning a mapping $g: \mathbb{R}^D \rightarrow \mathbb{R}^d$ that reduces dimensionality while preserving local relationships, the lower-dimensional manifold captures the essential structure of the high-dimensional data can be revealed.

StyleGAN, introduced by Karras et al. [52], revolutionised image synthesis by enabling fine-grained control over stylistic attributes through disentangled feature representations. Building upon Adaptive Instance Normalisation (AdaIN) [61], StyleGAN adaptively modulates feature maps to achieve precise style manipulation without relying on explicit content or style inputs. Instead of mapping the latent noise space $z \in \mathcal{Z}$ directly to the image space—as in standard GANs—a non-linear mapping network $f: \mathcal{Z} \rightarrow \mathcal{S}$ transforms z into a low-dimensional intermediate space $s \in \mathcal{S}$, which provides a more semantically meaningful and disentangled control space. Each dimension of the style vector s corresponds to interpretable attributes,

enabling independent control over visual factors such as shape, texture, and colour. Style mixing further allows hierarchical control across network layers, governing coarse and fine details separately. However, AdaIN's channel-wise normalisation introduced characteristic artifacts, which were addressed in StyleGAN2 [53] through Modulated Convolution (ModConv). By directly scaling convolutional weights w with s as $w'_{ijk} = s_i \cdot w_{ijk}$ and applying subsequent demodulation for normalisation, StyleGAN2 eliminates these artifacts while maintaining high-quality, scale-aware style control.

B. Generative Adaptable Segmentation Evolution (GASE)

The architecture of our GASE model is illustrated in Fig. 3. We denote the limited labelled dataset as $S = \{x, y\}$, where $x \in \mathbb{R}^{B \times H \times W \times 1}$ represents real images under distribution $P_{data}(x)$ with batch size B , $y \in \mathbb{R}^{B \times H \times W \times C}$ is the corresponding one-hot encoded ground-truth label batch with C classes. The G comprises two sub-components: the style mapping network $G_m: z \rightarrow s$, which maps Gaussian noise vectors $z \in \mathbb{R}^{1 \times L}$ to image style vectors $s \in \mathbb{R}^{1 \times P}$, and the synthesis generator $G_s: (y, s) \rightarrow \hat{x}$, responsible for synthesising diversified image $\hat{x}$ based on s drawn from a generated distribution $P_{fake}(x)$. The D , which processes both real x and synthesised $\hat{x}$, also comprises two sub-components: the segmentation branch $D_s: (\underline{x} = x \circ \hat{x}) \rightarrow \hat{y}$, which predicts the label $\hat{y} \in \mathbb{R}^{B \times H \times W \times C}$ for a mixed ($\circ$) input $\underline{x}$ between x and $\hat{x}$, and the confidence branch $D_c: (x, \hat{x}) \rightarrow (c_x, c_{\hat{x}})$, which generates individual confidence maps $c \in \mathbb{R}^{B \times H \times W \times 1}$ for both real and diversified samples.

C. Role of Generator, G

The G performs two tasks simultaneously: (1) Training Sample Diversification and (2) Style Representation Learning. For Task (1), the objective is to enrich the style of segmentation training samples by interpolating and diversifying the existing style features derived from inputs, while preserving the original semantic structures defined by the ground-truth labels. By exposing D to a broader range of style features that simulate

acquisition-level image variations in signal intensity and noise characteristics, G enhances segmentation robustness under such variability. For Task (2), the objective is to capture the semantic-dependent style features diversified by G during adversarial training and map them onto a low-dimensional, manifold-like representation for input interpretation, as illustrated in Fig. 2. To achieve this, we introduce a style-based G_s with a mapping network G_m inspired by StyleGANs [53] and further integrate it with Context Aggregation Module (CAM) [22] for a compact and efficient architecture.

Mapping network (G_m): In this work, we aim to construct a manifold-like representation of image style features from training samples and visualise it to provide an intuitive interpretation of complex input distributions. This is achieved using the mapping network (illustrated in Fig. 3(a)) proposed in StyleGANs for style-based learning (mentioned in Section III-A). Interpolations between two points along the geodesic in the s -space have been shown to produce smooth and semantically meaningful transitions in generated images [58]. Consequently, a manifold-like representation naturally emerges in s -space, capturing the style features learned by the D during segmentation training. Furthermore, spectral normalisation [62] is applied to all layers of the mapping network to ensure a smoother mapping between z and s along the manifold surface, as shown in Fig. 2, thereby mitigating sparsity. Additional implementation details are provided in the supplementary materials.

Style-based Generator (G_s): Inspired by StyleGANs, the ModConv operation [53], which directly modulates convolutional kernels for style injection, is integrated into our G to control image style based on the style vector. However, unlike StyleGANs—which reconstruct high-resolution images through progressive upsampling from a constant vector—our G synthesises images conditioned on the full-size ground-truth mask to preserve structural integrity. Thus, a mask-to-image translation architecture is needed to ensure that the input and output share identical spatial dimensions. Moreover, as illustrated in Fig. 1, unlike conventional GANs, where the D primarily serves as a variational loss for the G , the D in GASE simultaneously supervises G and performs semantic segmentation. Consequently, maintaining computational efficiency for both G and D , and balanced training dynamics between them, is crucial to prevent mode collapse—particularly since image synthesis tasks are inherently more difficult to stabilise than segmentation tasks.

Style-CAM: To address these challenges, our G is built upon CAM [22], a highly efficient and memory-compact CNN-based module originally designed for fast 3D medical image segmentation under resource constraints. By replacing sampling operations with dilated convolutions, CAM substantially improves computational efficiency by reducing expensive operations such as standard convolutions, while maintaining a large receptive field for feature extraction. This architecture’s inherent image-to-mask translation capability aligns naturally with our mask-to-image translation objective when inverted. However, the original ModConv operation is incompatible with dilated convolutions and cannot be directly implemented within the CAM framework. To overcome this limitation, we propose a new formulation of ModConv that modulates dilated convolutional kernels using the

style vector s , and integrate it with CAM as Style-CAM to inject style features on multiple scales. Style-CAM greatly expands the receptive field without upsampling for precise manipulation of style features across a wider spatial context conditioned on the input masks.

Since all layers within Style-CAM maintain a uniform spatial resolution due to the removal of sampling operations, multi-scale style mixing [53]—where distinct style vectors are applied to layers at different resolutions—is no longer practical. Therefore, a single style variable is used across all layers within Style-CAM to control global style variations. This design choice performs well in GASE, given the additional structural constraints provided by the input mask and the relatively consistent style characteristics present in MR images, compared to natural images. More details for our customised ModConv and Style-CAM are provided in supplementary materials.

Semantic Consistency: By conditioning GASE on the ground-truth label y , the goal is to preserve structure integrity of training samples during image diversification. Meanwhile, G is supervised by segmentation branch D_s to minimise the semantic difference between generated image $\hat{x}$ and the corresponding y input. This setup imposes semantic constraints on both input and output of GASE, guiding the mapping network to focus on learning manifold-like style representations conditioned on a fixed semantic structure during adversarial training. This confines G_m ’s learning space to style features only and further promotes faster convergence for G_m for balanced training dynamic. Once G is trained, various style vectors s , derived from the manifold constructed by G_m , are combined with the fixed ground-truth label y as input for G_s , generating diversified samples $\hat{x}$ that retain the semantic content of y while varying in style. By pairing $\hat{x}$ with y , we create synthetic labelled pairs $(\hat{x}, y)$ that match the label consistency of real image pairs (x, y) . This approach enriches the training sample pool without relying on artificial label maps commonly used in augmentation-based methods, which can lead to augmentation leakage [63].

D. Role of Discriminator, D

While supervising the G for image diversification, as in standard GANs, the proposed D is designed with two branches to perform complementary tasks (as illustrated in Fig. 3(c)): a segmentation branch D_s for the semantic segmentation task and a confidence branch D_c for confidence learning task.

Segmentation Branch (D_s) extracts features from the input images through an encoder and reconstructs predicted label maps using a decoder with skip connections, similar to the UNet architecture. During adversarial training, D_s is trained with a mixture of real and diversified labelled pairs synthesised by G for enhanced robustness against unseen acquisition-level image variations, meanwhile, supervising the semantic structure in the synthesised images $\hat{x}$ by back-propagating the segmentation loss to G , thereby enforcing semantic consistency between the $\hat{x}$ and their corresponding input masks.

Confidence Branch (D_c), on the other hand, performs pixel-wise confidence learning by generating a confidence map c that quantifies the similarity between the $\hat{x}$ and the original training

samples x . During training, multi-scale intermediate feature maps from the D 's encoder are aggregated to estimate pixel-level probability maps. Since this encoder is shared with the segmentation branch, its feature maps inherently capture semantic representations crucial for binary mask reconstruction. Consequently, the resulting confidence map reflects the style realism of each pixel relative to the underlying semantic structure within the input images. By back-propagating the confidence loss from D_c to the G during adversarial training, the mapping network within G is guided to construct a manifold-like representation of style features that compensates for the deficiencies identified by the confidence map. This process encourages the generated features to remain consistent with the anatomical boundaries defined by the input masks, thereby preventing global style variations (e.g., brightness or contrast changes) from distorting organ-specific structures, all while as an end-to-end framework.

E. Balanced Training Strategies

GAN-based architectures are notoriously difficult to stabilise during adversarial training and prone to mode collapse. This challenge is amplified in GASE, where both the G and D simultaneously perform complex tasks such as image synthesis and segmentation. Although efficient architectures have been selected to balance task complexity and convergence speed, additional stabilisation strategies are introduced to ensure robust and balanced training.

Diversification Regularisation (DR): Given the limited number of training samples, there is a risk of overfitting, where different noise vectors z map to similar style vectors s , producing identical synthetic images $\hat{x}$. To mitigate this, we introduce a Diversification Regularisation (DR) term into the G training to enhance feature diversity within $\hat{x}$, defined as:

$$\mathcal{L}_{Gp}(y, z_1, z_2; \theta_{G_m, G_s}) = \frac{\lambda_1}{\mathcal{L}_{mse}(\hat{x}_1, \hat{x}_2) + \lambda_2}, \quad (3)$$

where $\hat{x}_1$ and $\hat{x}_2$ are generated images from distinct noise inputs z_1 and z_2 , respectively. The reciprocal of the mean squared error penalises the model when similar images are produced from different noise inputs, encouraging the G to explore the input feature space and capture diverse image styles for manifold-like representations. The DR term further promotes interpolation between existing styles from limited training samples, enforcing smooth and continuous style transitions in the manifold shown in Fig. 2. The hyperparameters $\lambda_1 = 0.01$ and $\lambda_2 = 0.001$ were empirically chosen to cap the penalty and maintain a sharp gradient during back-propagation.

Real Synthetic Mixup (RSM): To better leverage diversified pairs $(\hat{x}, y)$ for more generalisable semantic segmentation while maintaining balanced adversarial training between the G and D , we propose the Real Synthetic Mixup (RSM). This module progressively introduces artificial signals into the training pipeline to prevent abrupt signal shifts during adversarial learning. As shown in Fig. 3(c), this is achieved by mixing signals from both real and synthetic labelled pairs when training D_s . Since both pairs share identical semantic structures, their style features are naturally blended for each anatomy, enhancing D_s 's

robustness to acquisition-level image variations by exposing it to a broader range of anatomical-constrained style features. RSM is inspired by the mixup augmentation technique [64], where x and $\hat{x}$ are blended using a weighting factor λ drawn from a Beta distribution with $\lambda = 0.2$. The mixed image $\underline{x}$ is defined as $\underline{x} = \lambda \cdot x + (1 - \lambda) \cdot \hat{x}$. To avoid destabilising training, the mixup operation is activated only once the synthetic distribution $P_{fake}(x)$ closely approximates the real data distribution $P_{data}(x)$ after a predefined number of epochs. The loss for D_s is defined as:

$$\mathcal{L}_{D_s}(x, \underline{x}; \theta_{D_s}) = \begin{cases} \mathcal{L}_{dsl}(D_s(x), y) & \text{if } e < T_m, \\ \mathcal{L}_{dsl}(D_s(\underline{x}), y) & \text{if } e > T_m. \end{cases} \quad (4)$$

where $\mathcal{L}_{dsl}$ is the weighted Dice Squared Loss (DSL) proposed in [22], e is the training epoches and T_m is predefined threshold to initialise RSM.

Random Gaussian Cutout (RGC): To further stabilise training and mitigate overfitting for segmentation task, we introduce Random Gaussian Cutout (RGC) layer—an adaptation of the Random Cutout layer [65] enhanced with Gaussian noise. As illustrated in Fig. 3(c), the RGC layer preprocesses the segmentation input by randomly replacing subregions of the input image with Gaussian noise. Unlike the original Cutout layer, which fills the removed regions with constant values, RGC employs Gaussian noise to enable smoother gradient flow between the two networks during back-propagation, thereby mitigating artefacts caused by abrupt signal changes introduced by the original cutout layer. In addition, RGC introduces extra randomness into D_s , reducing its dominance in supervising style features within $\hat{x}$ and shifting the primary style supervision to the confidence branch. More details of the RGC layer are provided in the supplementary materials.

F. Training Objective

During adversarial training, the G is trained solely based on implicit losses from the D , with style and semantic features in the generated image $\hat{x}$ supervised by the segmentation branch D_s and confidence branch D_c , respectively. To preserve authentic semantic information in $\hat{x}$, the loss for the predicted $\hat{y}$ is back-propagated to G as follows:

$$\mathcal{L}_{Gs}(y, z; \theta_{G_m, G_s}) = \mathcal{L}_{dsl}(D_s(\hat{x}), y). \quad (5)$$

Meanwhile, the style features of $\hat{x}$ are supervised by D_c through confidence learning. Unlike simple encoder in standard confidence networks with entire image as input, our D_c consists of multiple branches which receive multi-scale features from D_s 's encoder. This approach shifts D_c 's focus from global image realism to both local and global features derived from D_s 's perspective. Incorporating these shape-constrained features aligns D_c with our goal of constructing a segmentation-focused G , ensuring that the manifold built by G reflects valid input boundaries for segmentation. The confidence loss for training D_c is defined as:

$$\mathcal{L}_{Dc}(x, \hat{x}; \theta_{Dc}) = \mathcal{L}_{bce}(c_x, (1 - \zeta)) + \mathcal{L}_{bce}(c_{\hat{x}}, \zeta), \quad (6)$$

where $\mathcal{L}_{bce}$ is the binary cross-entropy loss, c_x and $c_{\hat{x}}$ are the confidence maps generated with x and $\hat{x}$, respectively, and ζ is

TABLE I

DATASET DISTRIBUTION ACROSS TESTING SCENARIOS INVOLVING STANDARD, PATIENT-LEVEL, SITE-LEVEL, AND SEQUENCE-LEVEL IMAGE VARIATIONS. PN = PELVIS (STANDARD VARIATION); PS = PELVIS (PATIENT-LEVEL VARIATION); MD = MSK-DESS; MS = MSK-SPACE; MF = MSK-FLASH; OD = OAIZIB-DESS

Scenario	Standard	Patient-level	Site-level	Sequence-level
Train Set	PN/MD	PN	MD	MD
Test Set	PN/MD	PS	OD	MS/ MF

a small Gaussian noise used to provide better gradient flow. The adversarial confidence loss for G is defined as:

$$\mathcal{L}_{Gc}(y, z; \theta_{G_m, G_s}) = \mathcal{L}_{focal}(c_{\hat{x}}, (1 - \zeta)), \quad (7)$$

where $\mathcal{L}_{focal}$ is the focal cross-entropy loss [66]. During the training, G and D will be trained iteratively for specified epochs. We define the hybrid loss function for G as:

$$\mathcal{L}_G(\theta_{G_m}, \theta_{G_s}) = \beta \mathcal{L}_{Gs} + \mathcal{L}_{Gc} + \mathcal{L}_{Gp}, \quad (8)$$

where β is set to 2 to enforce the semantic consistency. The hybrid loss function defined for D is:

$$\mathcal{L}_D(\theta_{D_s}, \theta_{D_c}) = \mathcal{L}_{Ds} + \mathcal{L}_{Dc}. \quad (9)$$

An abstract pseudo-code of GASE is presented in supplementary materials.

IV. EXPERIMENTS AND RESULTS

A. Experiment Setup

Since acquisition-level image variations are among the most frequently encountered by segmentation models during clinical deployment, three 3D MR datasets—Pelvis, MSK Knee, and OAIZIB Knee—with minor to moderate acquisition-level differences were selected to evaluate GASE’s robustness. As detailed in Table I, the datasets were divided into multiple scenarios to simulate different variation types: (1) Standard, with minimal data variation between training and testing sets; (2) Patient-level, caused by demographic differences within the same imaging data collection site; (3) Site-level, arising from different scanning protocols across different imaging data collection sites under the same contrast and sequence; and (4) Sequence-level, representing an extreme case involving different scanning sequences of the same contrast within the same imaging data collection. Visualisations of these variations are shown in Fig. 4. All datasets were standardised to a common spatial resolution before extracting 2D slices for evaluation. The training data in Table I were divided into three patient-level folds for cross-validation, and models trained on each fold were evaluated across all scenarios.

B. Dataset

(1) Open male pelvis dataset [67] comprises 211 3D MR examinations acquired from 38 patients, annotated with six classes: Image Background (BG), Urinary Bladder (UB), Body Multi-Tissue (BM), Regional Bones (RB), Rectum (RC), and Prostate Gland (PG). Twelve volumes exhibiting extreme variations in contrast and pixel intensity (Fig. 4(a)–(c)) are designated

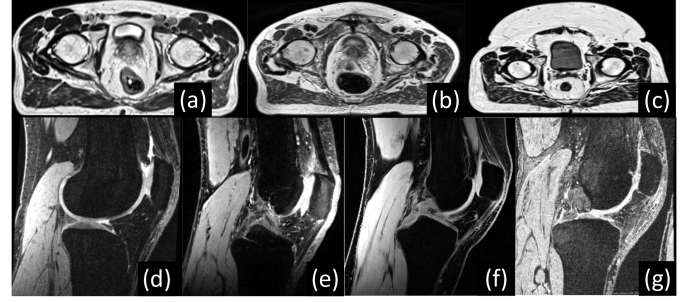


Fig. 4. Visualisation of acquisition-level variations in MR images. Figures (a–c) show axial pelvic T2-DESS images, illustrating patient-level differences: (a) normal, (b) full rectum, and (c) full urinary bladder with anterior fat tissues. Figures (d–g) show MSK sagittal knee images highlighting sequence-level differences—(d) DESS, (e) SPACE, (f) FLASH—and site-level differences between MSK and OAIZIB DESS acquisitions (d vs. g).

as patient-level variation testing cases, while the remaining volumes serve as “standard” cases for cross-validation. (2) Private MSK Knee dataset [68], [69] includes 3D MR knee examinations from 31 subjects across three sequences—DESS, SPACE, and FLASH. Six anatomical classes are defined for segmentation: Femoral Bone (FB), Femoral Cartilage (FC), Tibial Bone (TB), Tibial Cartilage (TC), Patellar Bone (PB), and Patellar Cartilage (PC). MSK-DESS is used for cross-validation, whereas MSK-SPACE (Fig. 4(e)) and MSK-FLASH (Fig. 4(f)) are reserved for sequence-level variation testing. Co-registered label maps from MSK-DESS are used for MSK-SPACE evaluation. (3) Open OAIZIB Knee dataset [70] contains 3D DESS MR knee examinations from 507 subjects and is used exclusively for site-level variation testing. Four classes are annotated: FB, FC, TB, and TC. To reduce computational cost, 51 subjects were randomly selected for patient-level variation testing. (See supplementary materials for additional dataset and preprocessing details).

C. Implementation

To evaluate GASE, we first compare it against supervised CNN models, including UNet (UN) [7] and Improved UNet (IUN) [10], a GAN-based model, ConfiGAN (CG) [44], and transformer-based models, SwinTransformer (ST) [71] and SwinUNetR (SUR) [72]. We also include NNUNet (NN) [9] for comparison, a non-baseline model with high scalability across imaging sequences by automating training pipelines for image segmentation tasks. Quantitative results are provided to compare GASE with other baseline models, and boxplots are presented to facilitate comparison between GASE and NNUNet due to their similar performance in some tests. Volumetric metrics (Dice Similarity Coefficient (DSC)) and surface-distance metrics (Hausdorff Distance (HD), Mean Surface Distance (MSD)) are used for assessment and tested for statistical significance using two-way/mixed ANOVA repeated (see supplementary materials). Models are selected based on the best validation loss under an 80–20 patient-level split, and all reported test results are averaged across three validation folds. Training is performed on an NVIDIA A100 GPU [73]. Further details, including

TABLE II

QUANTITATIVE COMPARISON ACROSS MODELS UNDER VARIOUS ACQUISITION-LEVEL IMAGE VARIATIONS. THE PELVIS DATASET INCLUDES FOUR CLASSES: URINARY BLADDER (UB), REGIONAL BONES (RB), PROSTATE GLAND (PG), AND RECTUM (RC). THE KNEE DATASET INCLUDES SIX CLASSES: FEMORAL BONE (FB), FEMORAL CARTILAGE (FC), TIBIAL BONE (TB), TIBIAL CARTILAGE (TC), PATELLAR BONE (PB), AND PATELLAR CARTILAGE (PC). EVALUATION IS PERFORMED USING THREE METRICS: DICE SIMILARITY COEFFICIENT (DSC), HAUSDORFF DISTANCE (HD) (MM), AND MEAN SURFACE DISTANCE (MSD) (MM). BOLD VALUES DENOTE THE BEST PERFORMANCE FOR EACH METRIC. THE MODELS COMPARED INCLUDE CONFI-GAN (CG), IMPROVED UNET (IUN), UNET (UN), SWINTRANSFORMER (ST), AND SWINUNETR (SUR). FULL RESULTS WITH STATISTICAL SIGNIFICANCE PROVIDED IN SUPPLEMENTARY MATERIALS.

		Knee-Related Experiments																		Pelvis-Related Experiments																	
		Standard (MSK-DESS)						Sequence-Level (MSK-PD)						Site-Level (OAIZIB-DESS)						Sequence-Level (MSK-FLASH)						Standard (Pelvis-Norm)						Patient-Level (Pelvis-Shift)					
	Metric	Our	CG	IUN	UN	ST	SUR	Our	CG	IUN	UN	ST	SUR	Our	CG	IUN	UN	ST	SUR	Our	CG	IUN	UN	ST	SUR	Our	CG	IUN	UN	ST	SUR						
FB	DSC	0.97	0.97	0.97	0.97	0.97	0.96	0.87	0.92	0.88	0.91	0.87	0.89	0.83	0.89	0.76	0.78	0.79	0.75	0.93	0.93	0.81	0.93	0.81	0.85	0.90	0.84	0.88	0.87	0.77	0.86						
	HD	10.8	6.60	9.85	17.2	6.51	19.8	21.7	11.4	55.0	35.1	27.4	56.3	44.0	14.8	61.3	47.0	42.0	71.2	19.4	22.2	82.8	32.9	63.2	76.2	7.74	13.8	9.49	12.8	19.7	25.9						
	MSD	1.23	1.02	1.37	1.74	1.04	1.99	3.84	2.22	10.1	4.39	5.94	10.9	9.60	3.49	15.4	10.3	9.61	17.4	2.37	3.56	19.7	4.81	14.9	18.9	2.27	4.49	2.51	4.32	9.22	3.92						
FC	DSC	0.90	0.85	0.88	0.88	0.85	0.86	0.85	0.75	0.77	0.80	0.63	0.71	0.77	0.62	0.68	0.72	0.51	0.63	0.83	0.76	0.80	0.79	0.78	0.75	0.93	0.90	0.91	0.91	0.89	0.89						
	HD	4.84	8.42	6.01	6.28	7.22	17.0	6.36	12.4	18.1	12.5	19.9	41.4	17.2	21.3	41.8	17.5	26.0	70.3	8.77	21.8	14.9	17.8	12.9	54.5	21.6	53.0	24.3	26.4	31.3	42.0						
	MSD	0.40	0.60	0.47	0.50	0.62	0.85	0.57	1.18	1.40	0.98	3.86	3.71	2.15	4.93	5.19	2.84	9.17	8.93	0.68	1.61	1.02	1.22	1.15	5.19	1.92	4.43	2.39	2.54	3.37	3.47						
TB	DSC	0.96	0.95	0.96	0.96	0.96	0.95	0.87	0.75	0.91	0.87	0.86	0.89	0.74	0.69	0.48	0.67	0.36	0.46	0.91	0.88	0.92	0.89	0.86	0.88	0.80	0.74	0.76	0.74	0.67	0.72						
	HD	9.97	6.80	7.91	13.2	6.18	16.2	60.2	27.8	38.3	50.4	23.0	47.1	47.4	29.1	69.6	51.9	44.8	81.4	62.0	42.9	51.4	40.6	37.5	71.2	7.92	9.71	7.86	12.4	16.6	17.7						
	MSD	1.39	1.22	1.14	1.57	1.03	1.85	12.3	6.21	5.88	8.87	4.61	6.37	11.5	10.0	18.8	12.9	17.4	23.1	9.34	7.45	7.34	5.26	6.38	11.3	2.93	3.49	3.24	6.56	10.6	4.90						
TC	DSC	0.83	0.78	0.83	0.82	0.78	0.81	0.77	0.65	0.73	0.73	0.52	0.67	0.72	0.54	0.65	0.65	0.43	0.59	0.80	0.74	0.77	0.75	0.72	0.76	0.94	0.88	0.91	0.91	0.82	0.88						
	HD	5.11	6.75	5.49	6.42	6.26	10.5	5.85	9.21	8.57	6.14	15.0	28.1	12.8	24.4	26.2	16.3	33.4	66.9	5.63	7.06	7.21	7.82	8.68	22.1	3.83	8.95	5.84	6.04	10.6	27.9						
	MSD	0.64	0.90	0.56	0.79	1.08	0.81	0.75	3.08	1.27	1.37	9.40	2.35	3.68	15.2	3.93	7.35	21.2	8.13	0.70	1.00	0.89	1.40	2.92	1.23	1.06	2.08	1.56	1.62	4.90	4.21						
PB	DSC	0.90	0.86	0.88	0.87	0.83	0.84	0.77	0.66	0.51	0.66	0.20	0.24	-	-	-	-	-	-	0.88	0.77	0.85	0.73	0.74	0.80	-	-	-	-	-	-						
	HD	4.66	4.34	4.90	7.51	9.90	15.5	10.4	14.1	28.2	30.4	89.9	70.7	-	-	-	-	-	-	14.2	10.2	11.3	46.2	25.0	67.5	-	-	-	-	-	-						
	MSD	1.00	1.61	1.21	2.52	4.16	2.89	4.74	8.26	9.40	12.6	79.4	21.4	-	-	-	-	-	-	2.59	5.23	3.80	15.6	19.7	9.51	-	-	-	-	-	-						
PC	DSC	0.89	0.83	0.86	0.85	0.80	0.82	0.71	0.56	0.57	0.46	0.26	0.45	-	-	-	-	-	-	0.83	0.67	0.80	0.71	0.64	0.74	-	-	-	-	-	-						
	HD	3.32	3.43	2.85	3.78	5.66	12.5	11.1	18.3	16.9	43.5	74.1	45.0	-	-	-	-	-	-	4.46	13.9	7.65	22.4	29.5	54.3	-	-	-	-	-	-						
	MSD	0.69	1.40	0.79	1.80	3.08	1.81	7.61	13.4	6.11	37.5	67.4	11.5	-	-	-	-	-	-	1.53	6.99	1.86	10.5	22.4	7.48	-	-	-	-	-	-						

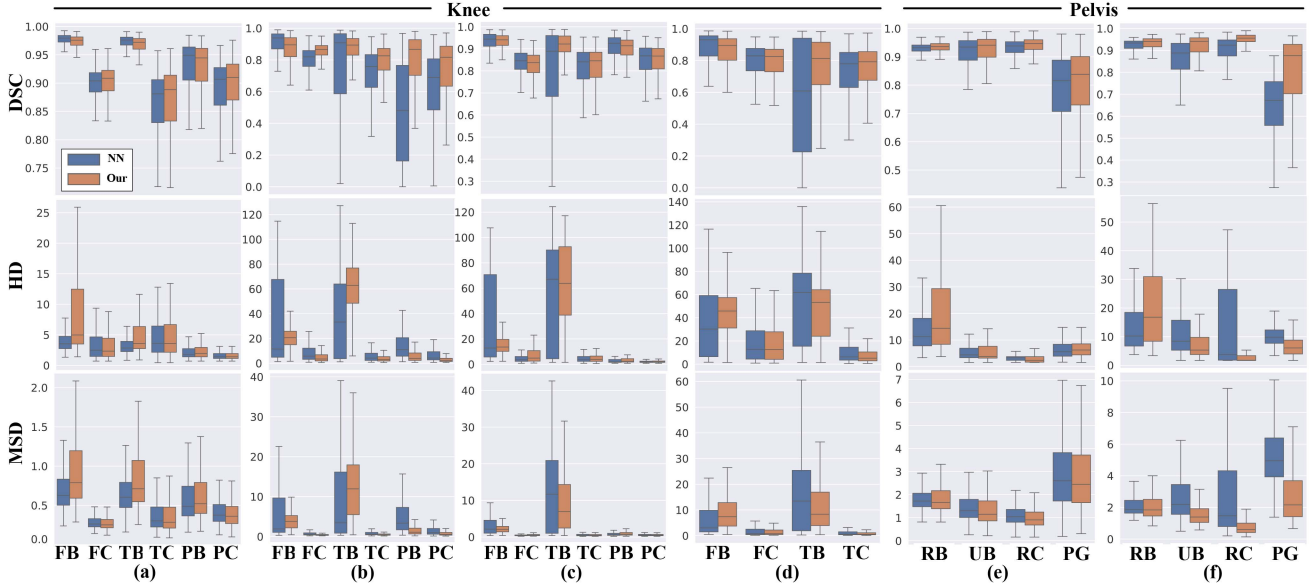


Fig. 5. Boxplot comparison of DSC, HD and MSD values between GASE (Our) and NNUNet for both knee and pelvis MR experiments across 3-fold cross-validation. (a) Standard test with MSK-DESS knee. (b-c) Sequence-level variation tests with MSK-SPACE and MSK-FLASH knee, respectively. (d) Site-level variation test with OAIZIB-DESS knee. (e) Standard test with pelvis data. (f) Patient-level variation test with pelvis data.

hyperparameters, loss functions, and runtime, are provided in the supplementary materials.

D. Comparisons With State-of-The-Art Methods

Standard tests: When minimal variations exist between the training and testing sets, GASE achieves strong segmentation performance even with limited training samples. As shown in Table II, CNN-based models—including GASE and NNUNet—generally outperform transformer-based counterparts, likely due to the lack of pre-training and the small dataset size. Among the CNN-based methods, GASE and NNUNet achieve the highest overall accuracy, with GASE showing superior performance on

imbalanced and more irregular structures such as cartilage in the knee dataset and the PG, RC, and UB in the pelvis dataset. Compared to rigid structures like bones, these irregular classes are more challenging to segment and often targeted for highly focused clinical analyses.

Patient-level tests: When models trained on the normal pelvis cases are tested on cases with patient-level variations, GASE is the only model that accurately segments imbalanced classes such as the PG, UB, and RC. It is apparent that NNUNet shows degraded performance in UB and PG segmentation, as illustrated in Fig. 5. Such outcome stem from patient-level variations observed in Fig. 4(c) (compared to Fig. 4(a)), including changes in UB contrast—likely due to varying bladder

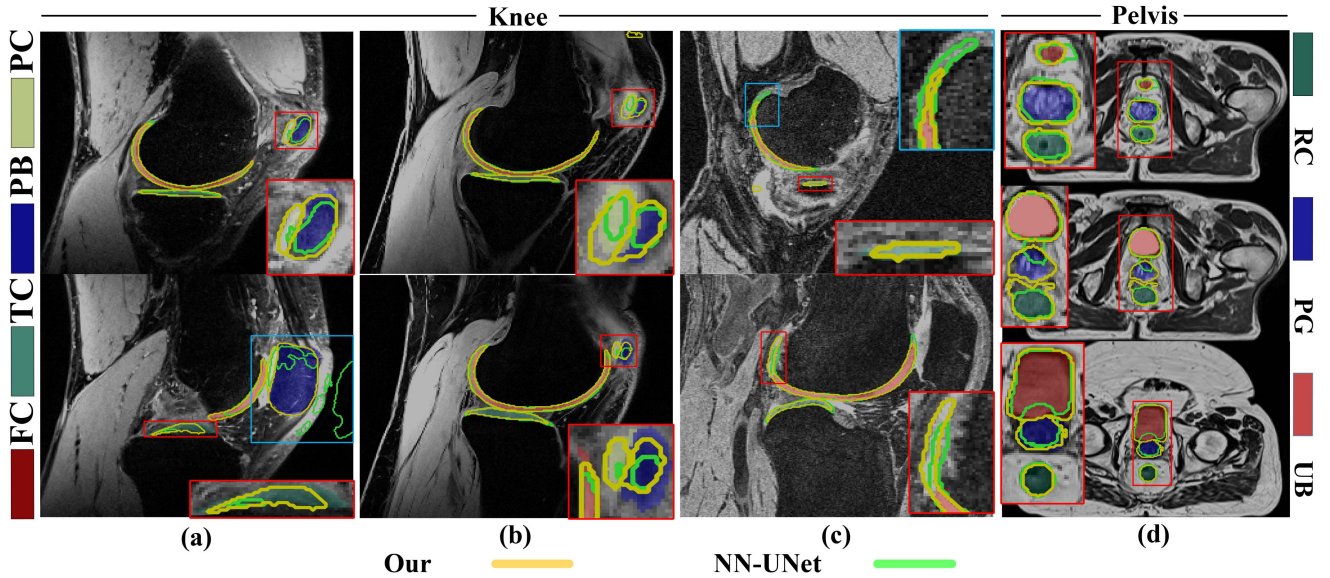


Fig. 6. Segmentation comparison between GASE (ours) and NNUNet for imbalanced classes, including Femoral Cartilage (FC), Tibial Cartilage (TC), Patellar Bone (PB), and Patellar Cartilage (PC) in the knee dataset, and Rectum (RC), Prostate Gland (PG), and Urinary Bladder (UB) in the pelvis dataset. Ground-truth labels are shown as solid color-filled regions, while predicted segmentations are outlined in gold for GASE and green for NNUNet. In panels (a–c), each column represents knee slices from the same patient: (a) sequence-level test with MSK-SPACE, (b) sequence-level test with MSK-FLASH, and (c) site-level test with OAIZIB. Panel (d) shows patient-level results for the pelvis dataset.

filling—and differences in surrounding fat tissue. As visualised in Fig. 6(d), GASE, unlike NNUNet, successfully delineates the interface between the PG and UB. This improvement arises from GASE’s ability to leverage diversified style features, enabling it to distinguish neighbouring classes even when similar or unseen features occur, whereas conventional CNNs may misclassify regions with overlapping feature patterns due to their inherent inductive biases and limited training samples.

Site-level tests: When GASE, trained on the MSK-DESS dataset, is evaluated on the unseen OAIZIB dataset—acquired under the same contrast and sequence but with different scanning protocols—it outperforms all other models and achieves performance comparable to NNUNet, showing slightly higher accuracy on imbalanced classes and significantly better results for TB. However, despite its superior overall performance, GASE still exhibits limitations due to substantial differences in image contrast and signal-to-noise ratios demonstrated in Fig. 4(g). As shown in Fig. 6(c), both NNUNet and GASE tend to over-segment cartilage, with NNUNet completely missing TC in the peripheral slices. These results highlight the need for additional standardisation when applying models to completely unseen datasets. Nevertheless, GASE demonstrates robustness to extreme feature variations and consistently outperforms competing models across imbalanced classes.

Sequence-level tests: Models trained on the MSK-DESS knee examinations are evaluated on other MSK sequences under the same contrast. As illustrated in Fig. 4, MSK-FLASH closely resembles MSK-DESS, whereas MSK-SPACE exhibits substantial contrast shifts, particularly around the patellar region. Consequently, while most models—including NNUNet—perform reasonably well on MSK-FLASH, only GASE maintains robust performance on MSK-SPACE. For instance, as shown in

Fig. 6(a), NNUNet fails to segment large portions of PB and PC, with parts of the background misclassified as PB due to contrast changes. It also misses a substantive portion of TC because of the considerable structural variations around the tibial spine region. In contrast, GASE effectively adapts to these variations by leveraging the diversified style representations of knee (illustrated in supplementary materials) similar to Fig. 2 during segmentation training, delivering consistently strong results—particularly for imbalanced classes—when generalizing across sequences with minor to moderate variations under the same dataset.

E. Style Representation Learning

The manifold-like representation of style features captured by the G after training on the pelvis dataset is illustrated in Fig. 2. To visualise this manifold, random noise vectors are first mapped to style vectors s using the mapping network and the resulting s are projected onto a 2D plane using t-SNE [74]. Synthetic images, generated by combining different s with a fixed label input through G , are then displayed along each s in the 2D plane for visual inspection. As shown, image samples along geodesic paths within such plane, while maintaining identical structural information, exhibit smooth and continuous transitions corresponding to interpretable semantic variations—such as changes in distribution of muscle and fat, signal intensity, and contrast. This behaviour indicates local smoothness and consistency in the learned latent representation, confirming the successful construction of a manifold-like surface that captures the distribution of structure-dependent style features from the training data. The observed smoothness further demonstrates GASE’s ability to interpolate new style features from existing

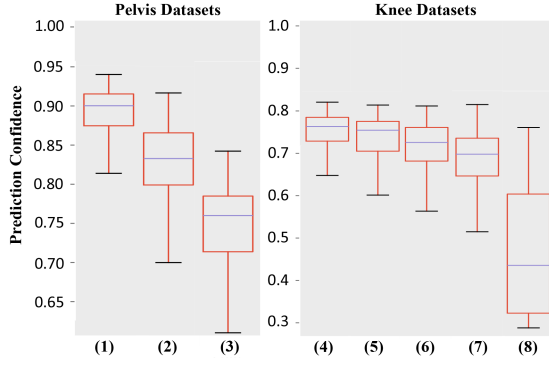


Fig. 7. Confidence score boxplots reflecting prediction reliability. The Pelvis-Norm-trained model is evaluated on: (1) Pelvis-Norm train, (2) Pelvis-Norm test, and (3) Pelvis with patient-level variation. The MSK-DESS-trained model is evaluated on: (4) MSK-DESS train, (5) MSK-DESS test, (6) MSK-SPACE, (7) MSK-FLASH, and (8) OAIZIB.

sparse ones—a crucial property for effective style diversification under data scarcity.

F. Model Interpretability:

Input Interpretation: As discussed in Section III-F, the style features within the generated images are supervised by the confidence branch D_c during confidence learning. Since D_c shares the same encoder as the segmentation branch D_s for feature extraction, the styles diversified by GASE—unlike those in conventional StyleGANs that aim to produce highly realistic images—are focused on enhancing the performance of the segmentation network. The manifold-like style representation captured by G_m after training therefore reflects the valid boundaries of the input style feature space as perceived by D_s . Visualising this representation provides deeper insight into how the segmentation model interprets different style variations, offering an interpretable view of the valid input distributions recognised by the network, as shown in Fig. 2. Further implementation details are provided in the supplementary materials.

Output Interpretation: After training, a confidence score is automatically computed as the average value of the confidence map produced by the confidence branch D_c , quantifying the similarity between the input image’s style and the style representation learned by the G . A higher score indicates higher similarity with the training samples in both style and semantic features and, consequently, greater prediction reliability, as the model is more likely to produce accurate results when the input samples are closer to the training distribution. Importantly, the confidence score does not measure segmentation accuracy directly; rather, it reflects the likelihood that a reliable mask has been generated for a given input. As shown in Fig. 7, the confidence scores align closely with dataset similarity relative to the training set: the training and testing sets achieve the highest scores, followed by the sequence-level variations (MSK-FLASH, MSK-SPACE), while the pelvis and OAIZIB datasets—representing patient- and site-level variations—exhibit the lowest confidence scores, consistent with the extreme variations observed in Fig. 4.

TABLE III

ABLATION STUDY FOR THE PROPOSED DESIGN CHOICES MENTIONED IN SECTION IV-G. CHECK MARKS INDICATE THE ACTIVE COMPONENTS FOR GASE VARIANTS. RESULTS BEFORE AND AFTER RSM THRESHOLD ARE REPORTED. BOLD INDICATE BEST RESULTS AFTER RSM. 1) FINAL GASE PERFORMANCE. 2) BACKBONE SEGMENTATION NETWORK PERFORMANCE. 3-4) MODEL STRUCTURE STUDY.

Index	Ds	Gs	Dc	Gm	Patient-level Variation Test (Pelvis)					
					PG		RC		UB	
					DSC	HD	DSC	HD	DSC	HD
					Befo. Aft.	Befo. Aft.	Befo. Aft.	Befo. Aft.	Befo. Aft.	Befo. Aft.
1	✓	✓	✓	✓	0.69 0.75	15.3 10.9	0.95 0.95	2.69 2.56	0.91 0.89	31.3 9.86
2	✓	-	-	-	- 0.71	- 20.2	- 0.93	- 10.5	- 0.80	- 16.4
3	✓	✓	-	✓	0.52 0.48	13.9 18.5	0.93 0.84	3.39 80.6	0.90 0.86	10.2 92.8
4	✓	✓	✓	-	0.69 0.73	9.29 12.2	0.94 0.94	4.18 3.49	0.89 0.87	35.0 21.6

G. Ablation Studies

Ablation studies have been conducted to evaluate the impact of the proposed design choices on final segmentation performance. Variants of GASE, with specific design elements selectively disabled, as detailed in Table III, were tested under the patient-level variations test using pelvis dataset.

Model Structure Study evaluates the key components of the GASE architecture—namely, the segmentation branch (D_s), confidence branch (D_c), synthesis generator (G_s), and manifold mapping network (G_m). Removing any of these elements greatly degrades performance, even leading to model collapse. Given D_s ’s central role in segmentation, we first test its standalone performance with other design choices disabled. As shown as in Study 2 in Table III, standalone D_s performs poorly under patient-level variations in the pelvis dataset, showing lower DSC for UB and PG and higher HD across all classes. Disabling the confidence branch D_c (Study 3) causes further degradation and unstable training, as the G receives no supervision over fine-grained style features. Removing G_m by replacing ModConv layers with standard convolutions (Study 4) turns G_s into a simple mask-to-image translation model without any style control, resulting in limited sample diversity and inferior HD, especially for the UB. Overall, these results demonstrate that each component is essential for enabling D_s to handle acquisition-level image variability. Without them, GASE loses its ability to synthesise meaningful style variations and cannot maintain robust segmentation performance, as further illustrated in the supplementary materials.

Manifold Construction Study evaluates the mechanisms that ensure successful image diversification and smooth manifold-like representation learning of style features in GASE, including the confidence branch (D_c), diversity regularisation loss (DR), context aggregation modules (CAM), and spectral normalisation within the mapping network (G_m). Disabling any of these components compromises the G ’s ability to produce high-quality synthetic samples $\hat{x}$, thereby degrading segmentation performance. Specifically, removing D_c eliminates fine structural details within $\hat{x}$ due to the lack of supervision, while omitting DR —which enforces distinct style generation—causes the model to produce identical images despite different noise inputs. Disabling Style-CAM reduces the receptive field of the modulated convolution kernels, limiting the model’s ability to

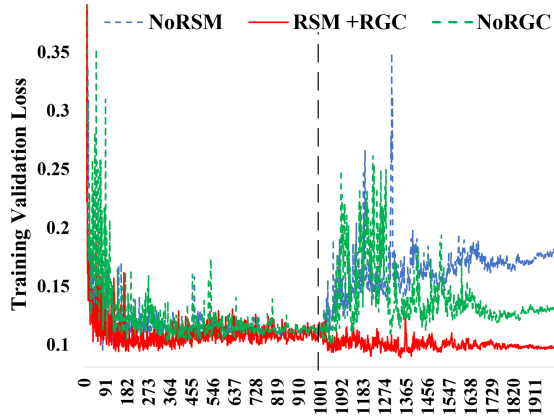


Fig. 8. Validation loss plot during GASE training with RSM enabled after 1000 epochs. The green and blue dotted lines show the training curves when RGC and RSM are disabled, respectively. The red curve, with both components enabled, results in the most stable training behavior and the lowest validation loss after introducing diversified samples.

synthesise realistic and spatially coherent style features. Finally, removing spectral normalisation disrupts the smoothness of the style manifold, fragmenting it into isolated clusters. Overall, inclusion of all mechanisms collectively enables GASE to map semantic-dependent style features onto a continuous manifold, facilitating stable diversification and anatomically consistent variations essential for robust segmentation. Visualisation results for these experiments are provided in supplementary materials.

Training Strategy Study investigates the strategies designed to enforce balanced training dynamics and improve segmentation accuracy with diversified samples, focusing on RGC and RSM. Although omitting these strategies does not cause model failure, it greatly reduces final accuracy and training stability. To disable RGC, full images are fed directly into the D without cropping, while disabling RSM uses synthetic images directly for training without real–synthetic mixing. As shown in Fig. 8, training becomes unstable, with a sharp increase in validation loss after the threshold (1000 epochs) when either RGC or RSM is removed. Without RGC, validation loss gradually increases after the threshold, suggesting overfitting, whereas disabling RSM leads to model collapse, indicating that raw synthetic signals are too intense for direct learning. Overall, the study confirms that RGC and RSM jointly enhance GASE’s stability and segmentation performance: RGC improves robustness and mitigates overfitting by injecting Gaussian noise during cutout operations, while RSM enables a smooth transition between real and synthetic domains, preventing abrupt signal changes and maintaining balanced learning between G and D . An additional ablation study comparing GASE with traditional augmentation techniques such as MixUp is provided in the supplementary materials. The results demonstrate that mixing generator-interpolated synthetic style features with real signals via the proposed RSM significantly enhances the robustness of the segmentation module to acquisition-level variations under extreme data scarcity, compared with the real-signal-only mixing strategy used in MixUp.

V. LIMITATION AND FUTURE WORK

One limitation of the current implementation is the absence of a path length regularisation term [53] during training, which may weaken implicit manifold regularisation and reduce global continuity of the manifold constructed. Nevertheless, our results indicate that local smoothness and consistency are largely preserved, suggesting the latent space remains well-conditioned in a local sense—likely due to the additional constraints imposed by semantic consistency and confidence learning. These mechanisms collectively stabilise the G by tightly constraining the learning space, thereby compensating, at least partially, for the lack of explicit regularisation. Future work involves adding path length regularisation term during model training may dramatically improve global continuity when capturing manifold-like style representation.

Another limitation lies in the suboptimal performance when segmenting large, rigid bone in the knee structures such as FB and TB in unseen sequences like MSK-SPACE and OAIZIB-DESS. GASE occasionally misses portions of the bones or misclassifies background regions as bone (as shown in the supplementary materials). Since this issue occurs exclusively in bone segmentation, it is likely attributed to the relatively uniform style features within bone regions across samples, which restrict the diversity of styles that the G can interpolate. Consequently, the model may confuse bone and background pixels due to their similar feature representations. Future work could address this by extending the framework to incorporate multiple style variables for independent anatomical style control, enabling finer-grained interpolation of local style variations and improved segmentation consistency across anatomical structures. Extending GASE to handle more extreme variations, such as cross-contrast or cross-modality adaptation, also represents a promising future direction to further exploit the advantages of the proposed framework.

One final limitation of the proposed framework is that it is currently implemented in 2D, processing individual slices rather than full volumetric data, to substantially reduce computational and memory demands during experimentation. In principle, the architecture is fully extensible to 3D by replacing all 2D operations with their 3D counterparts, which would allow better modelling of the inter-slice spatial continuity inherent in volumetric medical images. However, such an extension would inevitably increase memory consumption, model complexity, and training instability. As a compromise, 2.5D modelling strategies could be explored in future work to incorporate cross-slice contextual information (e.g., multi-view feature projection [29] or aggregation [75]) while maintaining lower computational cost and improved training stability compared to full 3D models.

VI. CONCLUSION

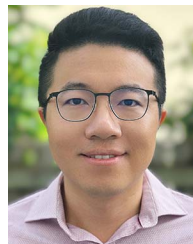
We introduced GASE, a robust and interpretable style-based generative framework designed to address the challenges of medical image semantic segmentation under data scarcity. The proposed framework enables the G to learn and exploit a manifold-like representation of structure-dependent style features derived from the input data, thereby enriching the

segmentation training samples through diverse yet anatomically consistent synthetic samples. The D , which simultaneously serves as the segmentation network, benefits from training on both real and stylistically diversified labelled data, substantially improving its robustness to acquisition-level image variations commonly encountered in clinical practice. Moreover, analysis of the learned style representation and the confidence map provides interpretable insights into the model's behaviour—defining the boundaries of valid input distributions and assessing prediction reliability. Extensive experiments on three distinct MR image datasets demonstrate that GASE achieves state-of-the-art image segmentation performance and exhibits superior generalisation to unseen acquisition-level variations, underscoring its potential for reliable and user-friendly clinical deployment.

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